

Start Early & Compounding

LESSON EXPLAINER · BEGINNER · GROWTH CAPITAL ACADEMY

What Is Compounding?

Compounding just means your money makes money — and then **that** money starts making money too. Imagine you invest \$100 and it grows by 10% in a year. You now have \$110. The next year, that 10% growth applies to the full \$110, not just your original \$100 — so you earn \$11 instead of \$10. It sounds small, but stretched over decades, that snowball effect becomes enormous.

Think of a snowball rolling down a long hill. At the top, it barely grows — there isn't much snow to stick to it yet. But the further it rolls, the bigger it gets, and the bigger it gets, the *faster* it picks up even more snow. Money invested early is that snowball right at the top of a very long hill.

Real Example: Sarah vs. Mike

Sarah starts investing \$200 a month at age 25. Mike waits ten years and starts investing \$400 a month — twice as much — at age 35. Both keep going until they retire at 65, and both earn a typical long-run average return of 7% a year.

	Sarah — starts at 25	Mike — starts at 35
Monthly investment	\$200	\$400
Years invested	40 years	30 years
Total contributed	\$96,000	\$144,000
Estimated value at 65	\$524,963	\$487,988

Even though Mike put in **50% more of his own money**, Sarah still ends up ahead — purely because her money had ten extra years to compound. That's the entire lesson in one example.

The Power of Reinvesting the Gains

A one-time \$10,000 investment growing at 7% a year, compared two ways: reinvesting each year's growth (compound) versus taking the same 7% only on your original \$10,000 every year (simple). Same rate, same starting amount — the only difference is whether the gains stay invested.

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 1	\$10,700	\$10,700	+\$0
Year 2	\$11,449	\$11,400	+\$49
Year 3	\$12,250	\$12,100	+\$150
Year 4	\$13,108	\$12,800	+\$308
Year 5	\$14,026	\$13,500	+\$526

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 6	\$15,007	\$14,200	+\$807
Year 7	\$16,058	\$14,900	+\$1,158
Year 8	\$17,182	\$15,600	+\$1,582
Year 9	\$18,385	\$16,300	+\$2,085
Year 10	\$19,672	\$17,000	+\$2,672
Year 11	\$21,049	\$17,700	+\$3,349
Year 12	\$22,522	\$18,400	+\$4,122
Year 13	\$24,098	\$19,100	+\$4,998
Year 14	\$25,785	\$19,800	+\$5,985
Year 15	\$27,590	\$20,500	+\$7,090
Year 16	\$29,522	\$21,200	+\$8,322
Year 17	\$31,588	\$21,900	+\$9,688
Year 18	\$33,799	\$22,600	+\$11,199
Year 19	\$36,165	\$23,300	+\$12,865
Year 20	\$38,697	\$24,000	+\$14,697
Year 21	\$41,406	\$24,700	+\$16,706
Year 22	\$44,304	\$25,400	+\$18,904
Year 23	\$47,405	\$26,100	+\$21,305
Year 24	\$50,724	\$26,800	+\$23,924
Year 25	\$54,274	\$27,500	+\$26,774
Year 26	\$58,074	\$28,200	+\$29,874
Year 27	\$62,139	\$28,900	+\$33,239
Year 28	\$66,488	\$29,600	+\$36,888
Year 29	\$71,143	\$30,300	+\$40,843
Year 30	\$76,123	\$31,000	+\$45,123

Based on a \$10,000 starting amount at a 7% annual return — the same illustrative figure used above, not a guarantee of future performance.

Worth knowing: the 7% figure above is a simplified, illustrative long-run average — real markets go up and down year to year, and past performance never guarantees future returns. The point isn't the exact number; it's the principle that time is one of the most powerful tools you have.

Quick Check: 5 Questions

Answer all five before turning the page to check yourself against the answer key.

1. What does "compounding" mean in investing?

- A. Your returns start earning their own returns over time
- B. Only your original deposits ever grow
- C. Your bank charges you extra interest on loans
- D. You must manually reinvest your dividends every single day

2. Why can starting 10 years earlier have such a big impact, even with smaller monthly contributions?

- A. Because markets always rise in an investor's first 10 years
- B. Because the extra decade gives compounding more time to work
- C. Because younger investors are charged lower fees by law
- D. It doesn't really matter — timing has no real effect

3. In the Sarah vs. Mike example, why does Sarah end up with more money despite contributing less overall?

- A. She chose better-performing investments than Mike
- B. Mike invested in a much riskier type of account
- C. Her contributions had 10 extra years to compound
- D. Sarah received a large one-off tax refund

4. Which statement best describes the "cost of waiting" to start investing?

- A. There's no real cost — you can always catch up later by investing more
- B. It only matters if you're investing more than \$1,000 a month
- C. Waiting is actually better because prices always fall over time
- D. Delaying reduces the time compounding has to work, which can be very hard to fully make up

5. What's the single biggest takeaway from this lesson?

- A. You need a large lump sum of cash to start investing
- B. You need to be a market expert before compounding helps you
- C. Time in the market is one of the most powerful tools for building wealth
- D. Compounding only works if you invest more than \$1,000 a month

Quiz Answer Key

- 1 **A** — Compounding means your returns start earning their own returns over time.
- 2 **B** — An extra decade gives compounding more time to work — that's why starting early matters so much.
- 3 **C** — Sarah ends up ahead because her contributions had 10 extra years to compound, not because of better investment choices.
- 4 **D** — Delaying investing reduces the time compounding has to work, which can be very hard to fully make up later.
- 5 **C** — Main takeaway: time in the market is one of the most powerful tools for building wealth.